Nonuniform Holomorphic Coordinate Compositions: Weighted Shadowing, Jet Budgets, and Image Exhaustion

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Abstract

We study the coordinate-composition stage of the supplied manuscript on uniformization of complete Kähler manifolds with strictly positive holomorphic bisectional curvature. The manifold, Ricci-flow, disc-envelope, and polynomial-model constructions are retained as inputs, not independently certified global theorems. We prove a backward shadowing estimate that retains the individual inverse Lipschitz factors and verifies every comparison inside its actual domain. Applied to the source's raw-orbit decay and macro grid, it replaces the sufficient choice $(d+1)\nu > 30$ by $5(d+1)\nu > 1$, with a strict margin and a subsequently chosen macro width. The resulting error decays faster than e^{-s}, which is enough for the source's separation, differential, and inverse-ball arguments. We also isolate a subsequential degree-to-logarithmic-radius-gap criterion for image exhaustion. Exact finite examples expose the need for amplification products, tube control, and a separate global degree bound. These are conditional quantitative refinements of this particular coordinate construction, not a new curvature theorem. Contribution assessment: C3/5, a reduced sufficient jet budget and explicit nonuniform domain-safe error bounds are proved while the global geometric inputs remain imported.

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1 Geometric problem and proof boundary

The supplied source [1] asserts that every connected, noncompact, complete smooth Kähler manifold (M, g) of complex dimension n with strictly positive holomorphic bisectional curvature is biholomorphic to $\mathbb{C}^n$. Strict positivity is pointwise; its statement assumes neither a global curvature upper bound nor a volume-growth or noncollapsing condition. The conclusion concerns the complex structure, not flatness of g. The global assertion is attributed to that manuscript throughout this report. Reading its proof text is not an independent certification of every analytic input or cited theorem.

The source normalizes the flow by $\partial_t h_t = -\operatorname{Ric}(h_t)$ and uses $\operatorname{Ric}(h) = -\partial\bar\partial \log\det h$. Fixing $o \in M$, its three clocks are physical time t, logarithmic time $\tau = \log t$, and

$$s = \rho(o,t), \qquad \rho = \log\frac{\det g}{\det h_t}, \qquad \frac{ds}{d\tau} = tR(o,t). \tag{1.1}$$

The s in the coordinate construction is not the physical-time variable also called s in the disc-variation section. All exponential rates below use the clock in (1.1). Cotangent contractions in the source use the inverse transpose metric, not a replacement Euclidean metric. Our vector and derivative norms are Euclidean only in the specified raw holomorphic coordinates, unitary at the flow-metric center. Conversion to the fixed physical metric g is explicitly retained.

There are no imposed boundary conditions on M. Auxiliary exhaustion problems do have boundary conditions. In particular, the source's Hodge heat approximations use its full zero-Dirichlet section convention; one cannot replace these by a free-space Green operator or a different absolute/relative Hodge boundary problem. None of our results makes such a replacement.

1.1 Architecture and retained obligations

All nine scientific modules were freshly displayed in full. Their roles are as follows; the listed global constructions remain source inputs here.

1. `preliminaries.tex`: weighted $\bar\partial$ tools, finite holomorphic separation and immersion, heat regularization, and a smooth distance exhaustion. Finite holomorphic separation is not properness.

2. `flow.tex`: localized Chern–Ricci approximations, scalar barriers and nonvanishing on compact sets, followed by exhaustion. Constants may depend on the fixed compact or localization parameters.

3. `discs.tex`: cotangent disc envelopes, compactness with an area bound, and holomorphic frames with controlled boundary seminorms.

4. `variation.tex`: almost-minimizing discs, deformations, negative-part removal, a backward clock, and a positive Schur-complement limit. Taylor radii may depend on the interior compact; the coefficient of the vanishing area penalty must not. The order of limits is substantive.

5. `ellipsoids.tex`: fiber optimization with the actual tangent–cotangent pairing, giving the joint plurisubharmonicity input.

6. `charts.tex`: Harnack consequences, completeness and simple connectivity, coherent inverse charts, and nonuniform contraction. These are not consequences of a local matrix inverse.

7. `dynamics.tex`: volume jets, polynomial automorphisms, rate flags, triangular regular steps, and payment for bad intervals. The forward degree bound holds on an unbounded subsequence, not on every index.

8. `uniformization.tex`: backward correction, separation, nonsingular differential, inverse balls, and a polynomial degree argument excluding a proper image. These are the stages refined below.

The introduction supplies the global statement and parameter map. External heat, factorization, regularity, and Liouville theorems referenced by these modules were not all independently audited. Our local theorems do not establish their hypotheses for an arbitrary positively curved manifold.

2 The source coordinate data

We state the imported data needed for an auditable conditional result. The source supplies local holomorphic maps $\Phi_s : B(r_0) \to M$, coherent inverse branches f_s on a compact exhaustion, and transitions $F_{s,u}$ fixing zero. They compose on their stated raw domains. The Φ_s need not be globally injective on their original domains. There is a common $0 < \sigma < 1$ such that, for each compact K,

$$\sup_K |f_s| \le e^{-\sigma s} \quad (s \text{ sufficiently large depending on } K). \tag{2.1}$$

For a permitted small $\gamma > 0$, the source also supplies

$$\sup_{|z| \le e^{-\gamma s}} |F_{s,u}(z)| \le e^{-c\gamma u}, \qquad u \ge s \ge s_\gamma, \tag{2.2}$$

where $0 < c \le 1$ after decreasing it. Choose, in this order,

$$0 < \gamma < \gamma' < \sigma/2, \qquad 0 < 10\nu < \min(c\gamma, \sigma, \gamma, 1). \tag{2.3}$$

The admissible range for γ is itself part of the chart input.

For each fixed integer $d \ge 2$ and sufficiently small fixed macro width $h > 0$, the polynomial-model proposition supplies a grid $s_k \to \infty$, polynomial automorphisms G_k of $\mathbb{C}^n$, and polynomials H_k, all fixing zero. Put $\Delta_k = s_{k+1} - s_k$ and $\delta = e^h - 1$. Eventually

$$1 \le \Delta_k, \qquad c_h\sqrt{s_k} \le \Delta_k \le \delta s_k. \tag{2.4}$$

The degrees of G_k, G_k^{-1}, H_k are uniformly bounded for fixed parameters. Writing $F_k = F_{s_k,s_{k+1}}$, their jets satisfy

$$J_d(H_{k+1} \circ F_k) = J_d(G_k \circ H_k). \tag{2.5}$$

The estimates needed below are

$$|H_{k+1}F_k(z) - G_kH_k(z)| \le e^{A_d\Delta_k + o(s_k)}|z|^{d+1}, \tag{2.6}$$

$$\|H_k\|_{\text{coeff}} \le e^{o(s_k)}, \qquad \|H_k'(0)^{-1}\| \le s_k^C, \tag{2.7}$$

$$\|(G_k^{-1})'(w)\| \le s_k^C e^{\Delta_k/2} + e^{C_d\Delta_k + o(s_k)}|w| \quad (|w| \text{ small}). \tag{2.8}$$

Fixed-degree constants have been absorbed into $A_d, C_d, o(s_k)$. Crucially A_d, C_d are chosen independently of h; other fixed parameters may affect the $o(s_k)$ term and the late starting index. Equation (2.6) concerns the two actual compositions, not an assumed analytic inverse of H_k on a uniform ball.

For either raw orbit $z_k = f_{s_k}(x)$ on a fixed compact, or $z_k = F_{s_a,s_k}(z)$ with $|z| \le e^{-\gamma s_a}$ and $k \ge a$, the source bounds imply, after a late start,

$$z_{k+1} = F_k z_k, \qquad |z_k| \le e^{-5\nu s_k}, \qquad \zeta_k = H_k z_k, \quad |\zeta_k| \le e^{-2\nu s_k}. \tag{2.9}$$

The second regime is uniform in sufficiently large a and its indicated initial ball; compact-dependent thresholds in the first regime are not claimed uniform over M.

Two further imported geometric facts are needed only for the passage to coordinates. On fixed compact sets, df_s has conorm at least $c_K e^{-s}$ in the fixed physical metric. The chart differential has a corresponding upper bound $C_K e^{s}$ while its image remains in a chosen slightly larger compact neighborhood. A first-exit argument is required to use that bound on a short raw segment. Finally, with

$$G_{k,l} = G_{l-1} \circ \cdots \circ G_k \quad (k < l), \qquad G_{k,k} = I, \tag{2.10}$$

the source supplies $\deg G_{0,k} \le D s_k$ on an unbounded subsequence. This degree statement is logically separate from (2.6).

3 Weighted backward shadowing with domains

The next result is a quantitative version of the source's finite-error sum. Its proof uses classical telescoping and holomorphic local uniform convergence. Keeping individual factors is useful when the common e^{Δ_k} majorant is very pessimistic.

Theorem 3.1 (Domain-safe product certificate)**.** *Let $G_i \in \mathrm{Aut}(\mathbb{C}^n)$ and let $\zeta_i : V \to \mathbb{C}^n$ be holomorphic, where V is a complex parameter domain. Suppose, uniformly on V,*

$$|\zeta_{i+1} - G_i \zeta_i| \le \varepsilon_i, \qquad |\zeta_i| \le Z_i, \qquad \mathrm{Lip}(G_i^{-1}|_{B(R_{i+1})}) \le L_i,$$

where $L_i > 0$. Define the positive products and tails

$$W_{k,j} = \prod_{i=k}^{j} L_i, \qquad E_k = \sum_{j=k}^{\infty} W_{k,j}\varepsilon_j < \infty. \tag{3.1}$$

Assume the actual tube condition

$$Z_{i+1} + \max(E_{i+1}, \varepsilon_i) < R_{i+1} \quad (i \ge k_0). \tag{3.2}$$

Then $y_k^{(l)} = G_{k,l}^{-1}\zeta_l$ converges uniformly on V, for each $k \ge k_0$, to a holomorphic ξ_k, with

$$|y_k^{(l)} - \zeta_k| \le \sum_{j=k}^{l-1} W_{k,j}\varepsilon_j, \qquad |\xi_k - \zeta_k| \le E_k, \tag{3.3}$$

$$|\xi_k - y_k^{(l)}| \le \sum_{j=l}^{\infty} W_{k,j}\varepsilon_j = W_{k,l-1}E_l, \qquad \xi_{k+1} = G_k\xi_k. \tag{3.4}$$

The empty product in the last formula is one when $l = k$. The same assertion holds locally uniformly if its bounds are available on each relatively compact parameter neighborhood.

Proof. For a fixed terminal l, set $y_l^{(l)} = \zeta_l$. At step i, the two arguments of G_i^{-1} are $y_{i+1}^{(l)}$ and $G_i\zeta_i$. By the induction hypothesis they are within E_{i+1} and ε_i, respectively, of ζ_{i+1}. Equation (3.2) puts both arguments and their joining segment inside $B(R_{i+1})$ before the inverse Lipschitz estimate is applied. Thus

$$|y_i^{(l)} - \zeta_i| \le L_i(|y_{i+1}^{(l)} - \zeta_{i+1}| + \varepsilon_i).$$

Backward induction proves the finite sum in (3.3). For consecutive horizons, at level l the difference is at most $L_l\varepsilon_l$; the same tube argument propagates it to at most $W_{k,l}\varepsilon_l$ at level k. Summing these

bounds proves the Cauchy property and (3.4). Uniform limits on compact sets are holomorphic. Passing through the fixed continuous G_k in the finite identities proves the recurrence. No derivative estimate outside its declared ball has entered the proof. □

Corollary 3.2 (Finite computation and a validated complement)**.** *If a bound $T_l \geq E_l$ is known, the finite pullback $y_k^{(l)}$ has error at most $W_{k,l-1}T_l$ from the selected infinite limit. If the terminal value is changed by at most η_l, and both resulting pullback chains and their comparison segments remain in the same balls, the additional error is at most $W_{k,l-1}\eta_l$. In particular the same selected limit is obtained if that product tends to zero. Bounds on a finite prefix alone do not supply T_l.*

Proof. The first assertion is (3.4); the second is successive Lipschitz comparison of the two chains. These comparisons have the explicit additional domain premise stated in the corollary. □

Example 3.3 (Weights cannot be deleted). On $\mathbb{C}$, take $G_i(z) = z/L_i$ with positive real L_i. For real defects $\varepsilon_i \geq 0$, define $\zeta_{i+1} = \zeta_i/L_i + \varepsilon_i$ on a finite chain. Direct substitution gives

$$G_{k,l}^{-1}\zeta_l - \zeta_k = \sum_{j=k}^{l-1} W_{k,j}\varepsilon_j.$$

Hence every weight in the estimate is attained, with no cancellation. For $G_i(z) = z/2$, $\zeta_i = 0$, infinitely many exact orbits $\xi_i = 2^{-i}a$ approach the zero orbit. Smallness alone does not select the limit; the terminal anchoring in Theorem 3.1 does.

Example 3.4 (A center derivative is not a tube bound). The polynomial automorphism $G(z, w) = (z, w + Az^2)$ has derivative I at zero. Its inverse maps $(t, 0)$ to $(t, -At^2)$, whose norm divided by $|(t, 0)|$ is $\sqrt{1 + |At|^2} > 1$ for $At \neq 0$. Thus the center inverse norm one cannot be used as a Lipschitz bound on an arbitrary ball. On $B(R)$ the elementary bound $1 + 2|A|R$ is valid, by differentiating G^{-1} and integrating on segments. This example uses a genuine global polynomial automorphism, not an inverse branch chosen across a singularity.

For a polynomial inverse $G_i^{-1} = A_i^{-1}z + \sum_{m=2}^{D_i} P_{im}(z)$, one usable local bound is

$$L_i(R) \leq \|A_i^{-1}\| + \sum_{m=2}^{D_i} mB_{im}R^{m-1}, \tag{3.5}$$

where $P_{im}(z) = \mathcal{B}_{im}(z, \ldots, z)$ and B_{im} bounds the operator norm of the actual symmetric multilinear map $\mathcal{B}_{im}$ in these coordinates. Differentiation proves (3.5); an unscaled list of coefficients is not itself this physical norm. Keeping such L_i in (3.1) may reduce a certified error. It does not assert measured computational speed or establish the infinite tail from finitely many coefficient evaluations.

4 A reduced sufficient jet budget

We now exploit the precise exponents in the source, rather than merely renaming a finite residual inequality. The amplification cost of backward iteration is exponential in s, so the decay exponent required by the physical chart conversion must be retained.

Lemma 4.1 (Exponential complement on the nonuniform grid)**.** *Suppose (2.4) holds, $L_i = e^{\Delta_i}$, and $\varepsilon_i \leq e^{-ps_i}$, where $p > 1 + \delta$. Set*

$$\kappa = p - 1 - \delta > 0, \qquad \beta = p - \delta = 1 + \kappa, \qquad C_\kappa = (1 - e^{-\kappa})^{-1}. \tag{4.1}$$

Then

$$E_k \le C_\kappa e^{-\beta s_k}, \tag{4.2}$$

$$\sum_{j=l}^{\infty} W_{k,j}\varepsilon_j \le C_\kappa e^{-s_k-\kappa s_l} \quad (l \ge k). \tag{4.3}$$

If $Z_i \le e^{-2\nu s_i}$ and $0 < \nu < 1/10$, the tube condition with $R_i = e^{-\nu s_i}$ holds at every sufficiently large index.

Proof. The product telescopes to $W_{k,j} = e^{s_{j+1}-s_k}$. Since $s_{j+1} \le (1+\delta)s_j$ and $s_j \ge s_l + j - l$ for $j \ge l$,

$$\sum_{j=l}^{\infty} e^{s_{j+1}-s_k-ps_j} \le e^{-s_k}\sum_{j=l}^{\infty} e^{-\kappa s_j} \le C_\kappa e^{-s_k-\kappa s_l}.$$

This proves both sums. The three quantities contributing to the next tube are bounded by $e^{-2\nu s_{i+1}}$, $C_\kappa e^{-\beta s_{i+1}}$, and $e^{-ps_{i+1}/(1+\delta)}$. Both β and $p/(1+\delta)$ exceed one, whereas $\nu < 1/10$. Their sum is eventually strictly smaller than $e^{-\nu s_{i+1}}$. This explicit check closes the finite-horizon induction; convergence was not assumed to prove domain validity. □

Theorem 4.2 (Conditional jet-order reduction)**.** *Assume all the source data in Section 2, including the availability of polynomial models for every fixed $d \ge 2$ and sufficiently small subsequently chosen h. The final coordinate construction can use any integer $d \ge 2$ satisfying*

$$\Lambda := 5(d+1)\nu > 1. \tag{4.4}$$

More explicitly choose $\delta = e^h - 1 > 0$ so small that

$$\delta < 1/10, \quad A_d\delta < 1, \quad C_d\delta < \nu/4, \quad (A_d+1)\delta < \Lambda - 1, \tag{4.5}$$

and fix

$$1 + \delta < p < \Lambda - A_d\delta. \tag{4.6}$$

After the remaining source parameters and a sufficiently late start are chosen, the corrected coordinates satisfy

$$|\xi_k - H_k z_k| \le C_\kappa e^{-\beta s_k}, \qquad \beta = p - \delta > 1, \tag{4.7}$$

in both source orbit regimes. The source's coherent injective map, nonsingular differential, compact forward decay, and inverse-ball inclusion follow with this error. The source degree input, or the alternative hypothesis of Theorem 5.2, is still required to conclude that its image is all of $\mathbb{C}^n$.

Proof. From (2.6) and (2.9),

$$|\zeta_{k+1} - G_k\zeta_k| \le \exp\bigl[-(\Lambda - A_d\delta)s_k + o(s_k)\bigr].$$

The strict upper inequality in (4.6) absorbs the $o(s_k)$ only after all parameters are fixed. Thus the defect is at most e^{-ps_k}. For $|w| \le e^{-\nu s_{k+1}}$, (2.8) gives

$$\|(G_k^{-1})'(w)\| \le s_k^C e^{\Delta_k/2} + \exp(C_d\Delta_k + o(s_k) - \nu s_{k+1}).$$

The first term is at most $e^{3\Delta_k/4}$ eventually because $\Delta_k \ge c_h\sqrt{s_k}$. The second tends to zero because $C_d\delta < \nu/4$. Their sum is at most e^{Δ_k} at a late index. The balls are convex, so the derivative bound gives the required Lipschitz estimate. Theorem 3.1 and Lemma 4.1 now give (4.7), holomorphic

limits, and exact recurrence. Notice that no estimate of H_k^{-1} on a uniform analytic domain was used.

Coherence and separation. On a fixed exhaustion neighborhood put $\Psi = G_{0,k}^{-1}\xi_k$ for a late fixed k. Recurrence makes this independent of k; coherence of the source inverse charts makes it independent of the neighborhood. If $\Psi(x) = \Psi(y)$, then

$$|H_k f_{s_k}(x) - H_k f_{s_k}(y)| \le 2C_\kappa e^{-\beta s_k}.$$

Write $H_k(z) = A_k z + N_k(z)$ with $A_k = H_k'(0)$. On the ball containing the two raw points, (2.1) and the fixed-degree coefficient bound give $\|DN_k\| \le e^{-\sigma s_k + o(s_k)}$. This is smaller than half the conorm s_k^{-C} of A_k. Integrating on the raw segment and using the reverse triangle inequality yields

$$|f_{s_k}(x) - f_{s_k}(y)| \le C s_k^C e^{-\beta s_k}.$$

Lift that segment by Φ_{s_k}. Until it first exits a fixed larger physical neighborhood its g-length is at most $C s_k^C e^{-(\beta-1)s_k} \to 0$. Choose the neighborhood to contain x, y with positive exit distance; the estimate prevents exit for large k. The lifted path connects x to y and its length tends to zero, so $x = y$. This step uses the source chart differential bound, not just the algebraic jet relation.

Nonsingular differential. On nested compact coordinate neighborhoods $U \Subset V \Subset M$, (4.7) holds uniformly on V. Cauchy estimates imply $\|d\xi_k - d(H_k f_{s_k})\| \le C_{U,V} e^{-\beta s_k}$ on U. The conorm of $d(H_k f_{s_k})$ there is at least $c s_k^{-C} e^{-s_k}$, using the previous bound for DH_k and the imported conorm of df_s. The ratio of the error to this conorm tends to zero since $\beta > 1$. Consequently $d\xi_k$, and hence $d\Psi$, is nonsingular. The injective map is a biholomorphism onto an open image Ω containing zero.

Forward decay. If $K \Subset \Omega$, its inverse image under this biholomorphism is compact. Since $G_{0,k}\Psi = \xi_k$,

$$\sup_K |G_{0,k}| \le e^{-\sigma s_k + o(s_k)} + C_\kappa e^{-\beta s_k} \le e^{-(\sigma/2)s_k} \tag{4.8}$$

eventually. Here $\beta > 1 > \sigma$ is sufficient.

Inverse balls. Fix large k and $r = e^{-\gamma s_k}$. The second orbit regime defines on $B(r)$ a holomorphic map

$$T_k = G_{0,k}\Psi\Phi_{s_k} = H_k + O(e^{-\beta s_k}).$$

The equality follows from the same coherent transitions, not from global injectivity of the original Φ_{s_k}. On $B(r/2)$ set $E_k^{\mathrm{loc}} = T_k - A_k z$. Fixed degree and Cauchy give

$$\|DE_k^{\mathrm{loc}}\| \le e^{-\gamma s_k + o(s_k)} + C e^{-(\beta-\gamma)s_k}.$$

Since $\beta > 1 > \gamma$, multiplication by $\|A_k^{-1}\| \le s_k^C$ makes this at most $1/4$ eventually. Moreover $E_k^{\mathrm{loc}}(0) = 0$. For $|w| \le e^{-\gamma' s_k}$ one has $|A_k^{-1} w| \le r/4$ at a late index, because $\gamma' > \gamma$. The map $z \mapsto A_k^{-1}(w - E_k^{\mathrm{loc}}(z))$ contracts the closed ball $\overline{B}(r/2)$ into itself: its norm there is at most $r/4 + r/8 < r/2$. Its fixed point solves $T_k(z) = w$. Thus

$$G_{0,k}^{-1} B(e^{-\gamma' s_k}) \subset \Omega. \tag{4.9}$$

This is a local contraction argument with a proved domain; the final global exhaustion requires the separate polynomial estimate below. □

Remark 4.3 (What was improved). The source already proves backward convergence and surjectivity. Its `uniformization.tex`, lines 11–107, chooses $(d+1)\nu > 30$, then a defect $e^{-10 s_k}$ and a final error $e^{-6 s_k}$. The improvement is the smaller sufficient jet budget (4.4) and its complete propagation through every subsequent comparison. This is not a claim that the threshold is optimal or that

lower degree necessarily costs less to compute. The source constants A_d, C_d, macro width, degree constant, and starting index are not effective uniformly in the geometry. In particular h cannot be chosen after observing an unspecified h-dependent error constant: the independence in (2.6) is indispensable.

Example 4.4 (An explicit conditional exponent allocation). Take the illustrative input values $\sigma = 4/5$, $c = 4/5$, $\gamma = 3/10$, $\gamma' = 7/20$, and $\nu = 1/100$, assuming this γ is in the imported permitted range. They satisfy (2.3). With $d = 20$, $A_d = 3$, $C_d = 2$, $\delta = 1/1000$, and $p = 51/50$, all inequalities (4.5)–(4.6) hold; in particular

$$\Lambda = 21/20, \quad \kappa = 19/1000, \quad \beta = 1019/1000.$$

The old displayed jet condition would require $d \geq 3000$ for this ν, whereas the new sufficient condition allows $d = 20$. These are consistent parameter fixtures, not constants certified for a specific curved manifold. The small positive κ entails a large tail factor C_κ and potentially a very late start. For example, the grid $s_k = (K + k)^2$ satisfies the upper spacing condition only after K is sufficiently large; the small macro width is not free.

5 Degree versus radius gap

For completeness we give the exact global algebraic argument and a conditional extension of its budget. It uses entire polynomial maps, not finite Taylor polynomials substituted for unknown entire maps.

Lemma 5.1 (Polynomial growth on concentric balls). *If $P : \mathbb{C}^n \to \mathbb{C}^n$ is polynomial of total degree at most m and $0 < R_0 < R_2$, then*

$$\sup_{B(R_2)} |P| \leq (R_2/R_0)^m \sup_{B(R_0)} |P|. \tag{5.1}$$

Proof. For a unit vector v and a unit scalar-valued linear functional ℓ, the function $p(z) = \ell(P(zv))$ is a polynomial of degree at most m. On the exterior of the radius-R_0 disc, $p(z)/z^m$ is holomorphic, including at infinity, and bounded by its boundary maximum. This gives the assertion along each complex radial line outside $B(R_0)$; inside that ball it is immediate. Take the supremum over v, ℓ. □

Theorem 5.2 (Subsequential gap criterion). *Let $\Omega \subset \mathbb{C}^n$ be an open set containing zero and let $P_j \in \mathrm{Aut}(\mathbb{C}^n)$ be polynomial, with degree m_j. Suppose real numbers $a_j > b_j$ satisfy, on a subsequence,*

$$\sup_K |P_j| \leq e^{-a_j} \quad \text{eventually for each } K \Subset \Omega, \qquad P_j^{-1} B(e^{-b_j}) \subset \Omega. \tag{5.2}$$

If $m_j/(a_j - b_j) \leq D < \infty$ on that subsequence, then $\Omega = \mathbb{C}^n$.

Proof. If Ω is proper, put $R_* = \mathrm{dist}(0, \mathbb{C}^n \setminus \Omega)$. It is finite and positive. Choose $R_0 < R_* < R_2$ so close that $D \log(R_2/R_0) < 1$. The compact ball $\overline{B}(R_0)$ lies in Ω. For every sufficiently late subsequence index, Lemma 5.1 implies

$$\sup_{B(R_2)} |P_j| \leq \exp[-a_j + m_j \log(R_2/R_0)] < e^{-b_j}.$$

The inverse inclusion in (5.2) therefore puts $B(R_2)$ inside Ω, contradicting the definition of R_*. No boundary regularity or attainment of a norm maximum on Ω is assumed. □

The source application is $P_j = G_{0,k_j}$, $a_j = (\sigma/2)s_{k_j}$, $b_j = \gamma' s_{k_j}$, and $m_j \le D_0 s_{k_j}$. Thus $D = D_0/(\sigma/2 - \gamma')$ suffices. More generally the gap can vary, and may be much smaller than s_k, provided the degree varies with it as stated. For example the abstract budgets $a_j = j^2 + 2j$, $b_j = j^2$, $m_j \le j$ satisfy the criterion with $D = 1/2$, although $(a_j - b_j)/a_j \to 0$. This only exhibits flexibility of the algebraic criterion; it does not produce new geometric charts.

Remark 5.3 (Limits of the criterion). Local nonsingularity does not imply global injectivity: $z \mapsto e^z$ has nonzero derivative and is not one-to-one or onto $\mathbb{C}$. Local injectivity and shadowing also do not supply a forward degree estimate. Conversely, finite degree by itself does not produce (5.2). Both compact forward smallness and the inverse-ball inclusion, with a positive gap, are necessary premises of this argument. A finite number of successful stages cannot certify an unbounded subsequence. The theorem is a conditional extension of the source's image-exhaustion step, not a relaxation of its curvature hypotheses.

6 Ram comparison, costs, and research directions

The fresh private readings are the original supplied `TR-core-pat-5.tex` and `stability.tex`, cited in [2, 3]. Their relevant proofs already distinguish specified physical norms from unscaled coefficients, finite inverses from infinite tail estimates, and local roots from unsupported global existence. The stability text, lines 221–230, explicitly gives the classical composition-error sum with products of downstream Lipschitz constants and requires both intermediate states to remain in valid sets. Lines 137–171 prove a contraction-ball certificate and its perturbation bound. We have not renamed those classical facts as new theorems of uniformization. The source-specific work here is the infinite backward specialization with explicit tubes, the source macro-grid summation, and the reduced jet order through the physical e^{-s} conorm threshold.

The core pattern's inverse-tail proof (lines 189–248) is a useful methodological comparison: order of multiplication, the actual norm, and the complement all matter. Its graph equations are not the nonautonomous polynomial dynamics here. Its finite matrix spectral arguments do not establish the source Harnack inequality or simple connectivity. The separately read metric and residual passages reinforce this boundary; no uninspected included private file is counted as read.

Advantages. Individual inverse factors retain nonuniform behavior. The tube test can reject an invalid use of an otherwise plausible small residual. The jet criterion isolates the exponent actually needed for geometric separation and differential invertibility. The horizon bound separates a computed finite composition from its assumed, or independently bounded, infinite complement. The degree-gap criterion permits nonconstant decay budgets.

Costs and limitations. Products can be large; replacing them by their individual sup bounds discards beneficial directional cancellation. The constant C_κ diverges as $\kappa \downarrow 0$. Reducing d can force smaller h, more stages, and later admissibility; no efficiency gain is claimed. Actual coefficients, rounding bounds, and global analytic tail bounds would be needed for a computer-assisted certificate on a given example. Our exact finite fixtures provide none of the source's infinite geometric estimates. The polynomial degree bound still depends on the source's rate-flag and bad-interval construction.

Further work. A useful next step is a certified anisotropic tube bound in the actual unitary chart metric, retaining ordered derivative products rather than only scalar Lipschitz products. Such a bound must control the nonlinear remainder on the full transported domain. An effective dependence of A_d, C_d and the late start on geometric data would turn the present asymptotic budget

into a usable finite stopping rule. A third direction is to estimate degree and radius gap together, instead of bounding each separately by s_k. None of these possibilities supplies missing Ricci-flow or global topology input.

7 Current literature and evidence scope

Public primary pages were checked on October 9, 2026 using public titles and URLs only. Bera–Verma [4] proves a uniform nonautonomous-basin result; its displayed hypothesis is a common ball and common constants $0 < A < B < 1$ with $A|z| \le |f_j(z)| \le B|z|$. We read the primary abstract, version record, and HTML introduction through its theorem and local jet discussion, not the complete proof. The nonuniform source bounds in Section 2 are not automatically those uniform hypotheses.

Datar–Pingali–Seshadri's April 2026 surface preprint [5] announces uniformization under positive real sectional curvature in complex dimension two; their June preprint [6] concerns a top Ricci integral and related algebraic conclusions. Wu's September 16 preprint [7] gives conditional uniformization of surfaces under strong Steinness and simple connectivity at infinity in the positive bisectional setting. For these three works we read primary abstracts and version metadata only. Their conditions and dimensions are not the unconditional all-dimensional assertion of the supplied manuscript. They are bibliographic comparators, not independent confirmations of it.

Every assigned public scientific module, wrapper, bibliography and README was displayed freshly in bounded untruncated intervals. The private stability file was read in full; the core-pattern file was read at lines 88–133, 189–248, 437–507, and 816–841 only. Exact complements and pre/post SHA256 records are in the accompanying read ledger. Reading the source bibliography does not mean reading its cited books and proofs. The captured public-tree identity is provenance, not a live branch check or theorem validation. No whole-corpus or independent global-proof audit is claimed.

The accompanying exact test script checks finite polynomial inverse and ordered composition identities, domain-safe finite shadowing, attained linear product weights, parameter inequalities, and negative controls. It uses rational arithmetic rather than sampling a curved manifold. The final owned run passed 184 exact parameter cases and 2,529 executed assertions, with zero floating-point cases. Repeated assertions within a finite trajectory are counted as assertions, not additional cases. The executable script hash is recorded in the final evidence package. Parent review and parent fixtures are excluded from owned totals.

8 Conclusion

The source's coordinate limit has three distinct obligations: correcting the local jet error inside valid shrinking domains, retaining enough decay after conversion to the physical metric, and excluding a proper global image by a forward polynomial degree bound. The product-tail certificate makes the first obligation explicit. The inequality $5(d+1)\nu > 1$, with all parameters chosen in the stated order, suffices for the second and preserves the source's inverse-ball conclusion. The degree-gap theorem states exactly what remains for the third. These results strengthen a local conditional stage without promoting finite algebra or a local inverse into a proof of global uniformization.

References

[1] OpenAI, *Uniformization of complete Kähler manifolds with positive bisectional curvature*, September 23, 2026, supplied cached preprint. Public source directory. All nine scientific modules read;

global inputs retained as described.

[2] M. Subbarao, supplied private original *TR-core-pat-5*, `RamPatsCore50-TRs/TR-core-pat-5`, file `TR-core-pat-5.tex`. Scoped metric, inverse-tail, residual, and nonlinear proof passages listed in Section 7. Confidential local source; not uploaded.

[3] M. Subbarao, supplied private original `RamNonlin-IE-PDE-astra/TR1-lin-nonlin`, file `stability.tex`, 295 lines. Physical scaling, finite perturbation, full nonlinear derivative, contraction neighborhoods, and composition-error proofs. Entire file read; its referenced private sources were not thereby read. Confidential local source; not uploaded.

[4] S. Bera and K. Verma, *Uniform non-autonomous basins of attraction*, arXiv:2211.15169v4, October 22, 2024. Primary abstract/version record and introductory theorem premises read; complete proof not independently audited.

[5] V. Datar, V. P. Pingali and H. Seshadri, *Uniformisation of complete Kähler surfaces with positive sectional curvature*, arXiv:2604.11220v1, April 13, 2026. Primary abstract and version record only.

[6] V. Datar, V. P. Pingali and H. Seshadri, *The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature*, arXiv:2606.19806v1, June 18, 2026. Primary abstract and version record only.

[7] J. Wu, *Conditional uniformization of Kähler surfaces*, arXiv:2609.18506v1, September 16, 2026. Primary abstract and version record only.